MSE-907 Capstone - Supervisor Feedback

Assessment: MSE907.3 - Final Research Project Report and presentation
Programme: Master of Software Engineering (Level 9) - Yoobee College of Creative Innovation
Student: Andre Luiz Santos Benedetti
Academic supervisor: Dr Helena Bahrami
Overall grade: A+
Source: assessment rubric Feedback (clean text extract; sectioned as in the original)

LO1: Literature Review (20%)

18.4 / 20 - A+

You have demonstrated excellent critical engagement with the literature concerning large language models in software engineering, AI-assisted code generation, developer productivity, software maintenance, and the limitations of automated coding tools. Your literature review in Section 2 is particularly strong because you distinguish between controlled coding benchmarks and the challenges encountered in real industry environments. You recognise that successful code generation on predefined programming tasks does not necessarily demonstrate reliable performance when requirements are incomplete, business rules are implicit, or developers must integrate changes into existing systems. Your review follows a documented search methodology informed by established systematic review guidance. You searched Scopus, IEEE Xplore, ACM Digital Library, and Google Scholar, initially retaining 22 sources and subsequently incorporating seven additional empirical studies through citation searching. Your critical comparison of studies reporting productivity improvements and studies identifying reduced productivity is particularly commendable. You appropriately recognise that the effectiveness of AI coding assistants depends on task complexity, developer experience, verification requirements, and the surrounding software development environment. Your identification of a research gap concerning AI-assisted development in existing business systems with stakeholder-owned rules provides a relevant justification for your investigation. Your distinction between code acceptance and code correctness is another significant strength. You recognise that accepting an AI-generated suggestion does not establish that it satisfies the intended business requirements. You could strengthen your literature review by developing a more systematic synthesis of the empirical findings across the reviewed studies, particularly concerning the relationships between task complexity, developer expertise, code acceptance, and subsequent rework. Although your review identifies relevant methodological differences, a structured comparison of experimental designs, participant characteristics, development environments, and evaluation metrics would help establish more clearly which findings are directly comparable. You should also distinguish more explicitly between research demonstrating measurable productivity effects and research investigating developer perceptions or code acceptance. These outcomes represent different aspects of AI-assisted development and should not be interpreted interchangeably. Your research-gap discussion could further clarify the extent to which your single-developer industry case contributes new empirical evidence rather than establishing general conclusions about AI-assisted software development. You could introduce a comparative evidence matrix covering controlled programming experiments, observational studies, developer surveys, and industrial case studies. For each study, you could identify the development environment, participant experience, task complexity, measurement method, observed outcomes, and limitations. You could then explain how your investigation contributes evidence about the acceptance, amendment, and rework of AI-generated code in a real business environment where some requirements were initially undocumented. This would strengthen the theoretical connection between the literature and your research questions.

LO3: Research-Informed Report (40%)

36 / 40 - A+

You have produced an excellent and technically substantial capstone research project demonstrating strong software engineering competence, independent problem-solving, and the ability to investigate AI-assisted development within a genuine industry environment. Your development of Park Tower Booking addresses a clearly defined operational problem at Crockers Property Group. Replacing duplicate booking entry across Palace CRM and an Excel spreadsheet with a centralised booking application represents a meaningful practical contribution. Your implementation of a Flask-based application integrated with Azure SQL, Microsoft Azure App Services, Microsoft Entra ID authentication, and the Palace REST API demonstrates strong competence in backend development, database integration, cloud deployment, and enterprise application design. Your application of a modular monolithic architecture is particularly appropriate for the project requirements. You explain how Flask blueprints, service layers, repositories, and data models separate business rules from database operations and user interface components. Your implementation of server-side booking validation, occupancy calculations, cleaning schedules, conflict detection, and role-based administration demonstrates good understanding of reliable software engineering principles. Your research methodology is particularly commendable because you separate the successful delivery of the application from the investigation of Cursor's contribution to the development process. Your analysis of 60 non-merge commits provides a documented basis for investigating AI-assisted development activities. You appropriately restrict conclusions about Cursor's involvement to the 24 commits for which supporting transcripts could be identified. Your findings that 18 of these 24 identifiable contributions were accepted and six were amended provide useful descriptive evidence about how the coding assistant was used. Your identification of five cases involving subsequent rework further strengthens the analysis. Your inclusion of a second independent coder is particularly valuable. You report agreement statistics and Cohen's kappa values rather than assuming that your original coding decisions are fully objective. Your discussion of the September booking-clearance error is an especially strong example of research-informed reflection. You demonstrate how an apparently plausible AI-generated implementation could violate an important business rule and why human review remained necessary. Your product validation is another significant strength. You report successful completion of 12 scripted scenarios and 15 user interface checks, supported by written acceptance from IT and Sales. Your System Usability Scale results, including a mean of 80.0 from three Sales participants and 97.5 from one IT participant, provide encouraging preliminary evidence of usability. Your explicit acknowledgement that the study cannot establish productivity improvements, and that the lines-of-code effort estimates are hypothetical rather than measured development time, demonstrates scientific integrity. You should strengthen the reliability of your empirical research by increasing the completeness of the development records used to identify Cursor's contributions. Although your analysis covers 60 commits, only 24 have sufficient transcript evidence to support classification of the assistant's involvement. Consequently, the reported acceptance and amendment proportions describe only the identifiable subset and cannot be generalised to all development activities. Your second-coder evaluation also requires further development. The reported task-type agreement of 58%, with Cohen's kappa of 0.47, indicates moderate agreement and suggests that some classification categories require clearer operational definitions. You could improve the coding framework by refining the distinctions between interface changes, integration work, defect correction, and business-rule modifications. Your software testing strategy is another important area for improvement. Although scripted acceptance testing provides useful evidence, the absence of an automated regression test suite creates a significant maintainability risk for a production application containing complex booking and cleaning rules. You should prioritise automated tests for occupancy calculations, date boundaries, booking conflicts, cleaning requirements, Palace synchronisation, and role-based access control. You could also strengthen the product evaluation by increasing the usability sample and conducting repeated testing under realistic operational

conditions. Your research would benefit from more detailed operational performance measurements, including response times, transaction failures, deployment reliability, and the frequency of booking corrections after deployment. You could implement an automated testing framework using pytest to verify critical booking rules. For example, a booking ending on 25 September should release the unit according to the agreed D+2 rule on 27 September. Automated tests could verify this behaviour across different dates, month boundaries, overlapping bookings, and cleaning conditions. For the research component, you could introduce a development logging protocol in which every AI-assisted task records the prompt, generated suggestion, human modification, final commit, and subsequent defect corrections. You could then repeat the coding analysis using a larger sample and calculate inter-rater agreement after refining the classification framework. A future controlled study could compare developers using Cursor with developers completing comparable tasks without an AI assistant. Measuring actual development time, code quality, defect frequency, and rework would provide stronger evidence about productivity and reliability. These improvements would strengthen both the engineering quality of the application and the scientific validity of the research conclusions.

LO4: Ethical and Cultural Issues (20%)

18 / 20 - A+

You have demonstrated a strong understanding of the ethical, legal, and cultural responsibilities associated with AI-assisted software development and the management of personal information within an industry application. Your discussion in Section 5 is particularly commendable because you connect ethical considerations directly to the actual development process, data structures, and operational limitations of Park Tower Booking. Your consideration of the New Zealand Privacy Act 2020 and Human Rights Act 1993 demonstrates appropriate awareness of the legal responsibilities associated with handling tenant information. You critically examine the collection of personal information, including names, contact details, school, country, and gender. Your recognition that the collection of certain demographic attributes requires a clearly documented operational purpose demonstrates an understanding of data minimisation and responsible information management. Your discussion of consent and your dual role as both application developer and researcher is another significant strength. You recognise that colleagues evaluating an application developed by someone they work with may provide overly favourable responses, and you explain the measures taken to reduce this potential bias. Your use of predefined acceptance criteria and an independent second coder demonstrates an effort to strengthen research integrity. Your consideration of AI-generated code and human accountability is particularly relevant. You explain why developers must verify generated implementations against actual business requirements rather than relying on apparently convincing code. Your cultural analysis is also commendable. You identify limitations in the application's handling of different naming conventions, including single names, compound surnames, family-name-first conventions, and Maori names containing macrons. Your acknowledgement that these limitations remain in the delivered application demonstrates transparency about the system's accessibility and inclusiveness. You also recognise the relevance of Maori data sovereignty and the need for appropriate consultation before introducing future demographic reporting involving Maori information. You could strengthen your ethical analysis by translating the identified privacy and cultural concerns into more explicit technical safeguards and organisational responsibilities. Although you identify the collection of school, country, and gender information as requiring further justification, you should establish a documented purpose for each field and determine whether its collection is necessary. You should also develop a formal data retention and deletion policy for locally stored booking information, activity logs, and records transferred to Palace. Your discussion of cultural inclusiveness could be strengthened through practical testing of international naming conventions and Unicode character handling across the complete application workflow. You should also consider the privacy implications of transferring personal information through external AI coding tools, particularly when debugging prompts or session transcripts may contain identifiable data. Your research would benefit from a more explicit explanation of how potentially sensitive

information was removed or anonymised before being submitted to Cursor. Finally, you should establish a clearer plan for addressing the accessibility limitations identified in your report, rather than leaving them solely as future recommendations. You could conduct a privacy impact assessment documenting the personal information collected, its operational purpose, storage location, retention period, access permissions, and disclosure requirements. For cultural inclusiveness, you could create a test dataset containing representative naming structures, including single names, compound surnames, family-name-first conventions, and names containing Maori macrons. You could then verify that names are correctly preserved through form submission, database storage, Palace integration, and exported reports. You could also establish a development policy requiring the removal of identifiable tenant information before any debugging material is submitted to an external AI coding assistant. These improvements would provide stronger evidence that privacy, ethical responsibility, and cultural inclusiveness are embedded in the system's engineering practices.

LO5: Presentation (20%)

19 / 20 - A+

You delivered an excellent, professional, and well-structured presentation that effectively communicated your industry project, research methodology, software implementation, empirical findings, and conclusions. Your presentation was particularly effective in distinguishing between the operational problem addressed by Park Tower Booking and the academic investigation of AI-assisted software development. Your introduction on Slide 2 clearly established the business problem of duplicate booking records and explained the practical value of replacing the spreadsheet-based process with an integrated application. Your discussion of the development process on Slide 3 demonstrated a clear understanding of the respective responsibilities of Cursor, the developer, and the IT team. Your explanation of the September booking-clearance error provided a particularly effective example of why human verification remains essential when using AI-generated code. Your presentation of the research questions on Slide 4 was clear and academically focused. You appropriately distinguished the investigation of Cursor's development contributions from the separate validation of the delivered application. Your presentation of the commit analysis on Slide 5 was particularly commendable. You communicated the distinction between identifiable and unclassified AI contributions and presented the acceptance, amendment, and rework findings transparently. Your discussion of second-coder agreement further demonstrated awareness of the importance of research reliability. Your explanation of human control points on Slide 6 effectively connected the research findings to practical software engineering responsibilities, including requirements definition, code review, server-side validation, IT oversight, and business acceptance. Your presentation of product validation on Slide 7 clearly communicated the successful scripted tests, user interface checks, usability scores, and written acceptance evidence. Your proposed future research design on Slide 8 was another strength. You demonstrated an understanding of how a larger controlled experiment could overcome limitations associated with a single-developer observational study. Your application screenshots on Slides 10-17 provided valuable evidence of the completed software solution and effectively demonstrated the booking, pipeline, cleaning, conflict management, and administration functions. You presented your research confidently and responded effectively to the evaluators' questions, demonstrating strong technical understanding and the ability to explain your research decisions and limitations. You could further strengthen your presentation by simplifying some of the detailed research findings and providing a more concise visual explanation of the relationship between AI-generated code, human review, and subsequent rework. For example, the commit analysis could be presented as a flow diagram showing the 60 analysed commits, the 24 identifiable Cursor contributions, and their associated outcomes. You could also provide greater emphasis on the implications of the second-coder agreement results, particularly the limitations associated with moderate agreement on task classification. Your presentation would benefit from a clearer visual summary of the limitations of the product evaluation, including the small usability sample and the absence of automated regression testing. You could also make the distinction between the production application and the development environment used for scripted validation more prominent. Finally, you could strengthen the conclusion by clearly separating what the research demonstrates about Cursor's use from what remains unknown about its effect on developer productivity. You could introduce a final research findings slide with three sections: observed AI-assisted development outcomes, demonstrated human control mechanisms, and limitations of the available evidence. For example, you could explain that 75% acceptance among identifiable Cursor contributions demonstrates the frequency of accepted suggestions within the observed subset, but does not establish a 75% correctness rate or prove a productivity improvement. This would make the scientific contribution and limitations of your research particularly clear to the audience.

Feedback to Learner

Overall: A+ (92 / 100)

You have produced an excellent postgraduate capstone research project demonstrating strong software engineering competence, independent research capability, and the ability to investigate AI-assisted development within a real industry environment. Your work is particularly commendable for successfully delivering Park Tower Booking, a functional application addressing a genuine operational problem at Crockers Property Group. Your implementation demonstrates effective integration of backend development, cloud infrastructure, enterprise authentication, database management, external API communication, and business-rule validation. Your research contribution is strengthened by the systematic analysis of development records, the use of supporting Cursor transcripts, the inclusion of a second coder, and the investigation of human oversight during AI-assisted development. Your distinction between successful software delivery and evidence of AI productivity improvements is particularly important. You demonstrate appropriate scientific caution by avoiding unsupported claims about development speed or code correctness. Your discussion of the September booking-clearance error provides a meaningful example of the limitations of AI-generated code when requirements depend on undocumented business knowledge. Your ethical analysis is another strength, particularly your recognition of personal information management, researcher bias, international naming conventions, and the responsibilities associated with AI-assisted development. Your presentation was professional, clearly organised, and confidently delivered. You successfully communicated both the practical implementation and academic investigation and responded effectively to the evaluators' questions. To strengthen your work further, you should prioritise automated regression testing, more complete collection of AI-assisted development evidence, improved coding reliability, and expanded product evaluation. You should also address the outstanding privacy and cultural accessibility concerns identified in your report, particularly the justification for demographic data collection, retention policies, and support for diverse naming conventions. Overall, you have demonstrated excellent achievement at Master's Level 9, with particular strengths in industry software development, empirical research, critical reflection, human oversight of AI-generated code, and professional academic communication.